# MZUZU UNIVERSITY

## Department of Mathematics

## EMT 1101 - Precalculus

## End of Semester Supplementary Examination

**Date: Some day in 2016** **Time: Some time $x$ (3 hours)**

1. Check that the paper has **14 questions**.
2. Attempt all questions
3. Write answers in the space provided
4. Use proper notation
5. Simplify your answers as far as possible
6. **No Calculators**

Name _____

Reg Number , Programme: _____

Professor's / Lecturer's name: _____

1. Rationalize the denominator and simplify $\dfrac{\sqrt{x}+\sqrt{2}}{\sqrt{x}-\sqrt{2}}$ (3)

**Solution:** $\frac{\sqrt{x}+\sqrt{2}}{\sqrt{x}-\sqrt{2}} = \frac{\sqrt{x}+\sqrt{2}}{\sqrt{x}-\sqrt{2}} \times \frac{\sqrt{x}+\sqrt{2}}{\sqrt{x}+\sqrt{2}} = \frac{x+2\sqrt{2x}+2}{x-2}$

2. Factor the expression $y^4 - (p+q)y^3 + (p^2q + pq^2)y - p^2q^2$ (4)

**Solution:** $y^4 + (p+q)y^3 - (p^2q + pq^2)y - p^2q^2 = y^4 - y^3q - y^3p + y^2pq - y^2pq + pq^2y + pq^2y - p^2q^2$
$= (y^2 - pq)(y^2 - yq - yp + pq)$
$= (y^2 - pq)(y - p)(y - q)$

3. (a) Solve the equation $x(2x - 13) = -6$ (4)

**Solution:** Remove bracket and re-arrange the equation $2x^2 - 13x + 6 = 0$
$(2x - 1)(x - 6) = 0$
$x = \frac{1}{2}$ or $x = 6$

(b) The vertices of triangle ABC are $A(1, 1)$, $B(9, 3)$ and $C(3, 5)$.Find the perimeter of the triangle formed after joining the midpoints of the three sides of triangle ABC and leave answer as an irrational number. (5)

**Solution:** Find mid points $AB = (\frac{10}{2}, \frac{4}{2}) = (5, 2)$ $BC = (6, 4)$ $AC = (2, 3)$ New triangle with points $P(5, 2)$, $Q(6, 4)$ and $R(2, 3)$ is triangle PQR Find distance $PQ = \sqrt{5}$, $QR = \sqrt{17}$ and $PR = \sqrt{10}$ So perimeter of triangle PQR= $\sqrt{5} + \sqrt{17} + \sqrt{10}$

4. (a) A line with a slope of $-3$ passes through the point $(3, 6)$. Find the area of the triangle in the first quadrant formed by this line and the coordinate axes. (4)

**Solution:** Using point slope form, the equation is $y - 6 = -3(x - 3)$
$y = -3x + 15$, so $y$-intercept is 15 and $x$-intercept is 5. Area of this triangle is $\frac{1}{2} \times 15 \times 5 = \frac{75}{2} = 37.5$ square units.

(b) Determine the center and radius for the circle $3x^2 + x + 3y^2 + 3y - 1 = 0$ (5)

**Solution:** Divide by 3 throughout the equation $x^2 + \frac{x}{3} + y^2 + y - \frac{1}{3} = 0$
Using complete the square we get $\left(x + \frac{1}{6}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{1}{3} + \frac{1}{36} + \frac{1}{4} = \frac{11}{18}$
center is $\left(-\frac{1}{6}, -\frac{1}{2}\right)$ and radius is $\sqrt{\frac{11}{18}}$

5. Determine a quadratic equation in the form $ax^2 + bx + c = 0$ with roots $r_1 = \frac{1}{3}\left(4 - \sqrt{5}\right)$ and $r_2 = \frac{1}{3}\left(4 + \sqrt{5}\right)$ (6)

**Solution:** $b = -(r_1 + r_2) = -\left(\frac{4}{3} - \frac{\sqrt{5}}{3} + \frac{4}{3} + \frac{\sqrt{5}}{3}\right) = -\frac{4}{3}$
$c = r_1 r_2 = \frac{1}{3}\left(4 - \sqrt{5}\right) \times \frac{1}{3}\left(4 + \sqrt{5}\right) = \frac{1}{9}\left(16 - 5\right) = \frac{11}{5}$
Therefore the equation is $x^2 - \frac{8}{3}x + \frac{11}{9} = 0 \Longrightarrow 9x^2 - 24x + 11 = 0$

6. Determine all real number solutions of $\sqrt{2 - x} - 10 = x$ (4)

**Solution:** $\sqrt{2 - x} - 10 = x$
$\sqrt{2 - x} = x + 10$
$2 - x = (x + 10)^2$
$2 - x = x^2 + 20x + 100$
$x^2 + 21x + 98 = 0$
$(x + 7)(x + 14) = 0$
Either $x = -7$ or $x = -21$
Upon checking, $-21$ is an extraneous root, so $x = -7$ is the solution

7. A piece of wire $6x$ centimetres long is bent into an equilateral triangle. For which values of $x$ will the area be numerically less than the perimeter? (7)

**Solution:** Perimeter $= 6x$
Area $= \frac{1}{2}bh = \frac{1}{2}2x \times \frac{x\sqrt{3}}{2} = \frac{x^2\sqrt{3}}{2}$

This yields the inequality $\dfrac{x^2\sqrt{3}}{2} - 6x < 0$
$x^2\sqrt{3} - 12x < 0$
$x(x\sqrt{3} - 12) < 0$

The key values are $x = 0$ and $x = \frac{12}{\sqrt{3}} = 4\sqrt{3}$

| Interval | Test No. | $x$ | $x\sqrt{3} - 12$ | $x(x\sqrt{3} - 12)$ |
|---|---|---|---|---|
| $(-\infty, 0)$ | -1 | - | - | + |
| $(0, 4\sqrt{3})$ | 2 | + | - | - |
| $(4\sqrt{3}, \infty)$ | 8 | + | + | + |

This yields $x \in (0, 4\sqrt{3})$

8. Consider the function $g(x) = x^2 + 3x$.
   (a) Find and simplify the average rate of change of $f$ over the interval $[a, 5]$ (4)

**Solution:** The rate of change is $\frac{f(5)-f(a)}{5-a} = \frac{40-a^2-3a}{5-a} = \frac{(a+8)(a-5)}{a-5} = a+8$

(b) Use the result in part a) to find the average rate of change of $f$ over the interval $[-2, 5]$ (2)

**Solution:** Let $a = -2$, the the average rate of change is $a + 8 = -2 + 8 = 6$

9. Let $g(x) = \sqrt{1-x^2} - 3$ and $f(x) = x - 4$.

(a) Use translations to sketch the graph of $g(x)$ (3)

**Solution:** Shift the graph of $y = \sqrt{1-x^2}$ three places down

(b) Specify the domain and range of $g(x)$ (2)

**Solution:** The domain is $[-1, 1]$ and the range is $[-3, -2]$

(c) Evaluate $g \circ f$. (3)

**Solution:** $g \circ f = g(x-4) = \sqrt{1-(x-4)^2} - 3 = \sqrt{-x^2+8x-15} - 3$

(d) By considering the domain of $g(x)$, specify the domain of $g \circ f$ (2)

**Solution:** The domain is the set of all values $x$ such that $-1 \leq f(x) \leq 1$. i.e. $-1 \leq x - 4 \leq 1$ from which $3 \leq x \leq 5$, so the domain of $g \circ f$ is $[3, 5]$

(e) Considering the definition of $g \circ f$, explain how you would sketch the graph of $g \circ f$. (2)

**Solution:** Since $g \circ f = g(x-4)$, then the graph of $g \circ f$ is obtained by shifting the graph of $g$ four places to the right.

10. Suppose that a basket weaver knows that the daily production cost to build $x$ baskets is given by the function $C$, where

$$C(x) = 1000 + 900x - 10x^2$$

That is, $C(x)$ represents the cost in Malawi Kwacha of weaving $x$ baskets. Furthermore, suppose that the number of baskets that can be woven in $t$ hours is given by the function $f$, where

$$x = f(t) = 5t$$

(a) Compute $(C \circ f)(t)$. (3)

**Solution:** $C(f(t)) = 1000 + 900(5t) - 10 \times (5t)^2 = 1000 + 4500t - 250t^2$

(b) Compute the production cost on a day that the the basket weaver works for 3 hours. (2)

**Solution:** $C(f(3)) = 1000 + 4500 \times 3 - 250 \times 3^2 = K12,250$

(c) If the basket weaver works for 6 hours instead of 3 hours, is the cost twice as much? (3)

**Solution:** $C(f(6)) = 1000 + 4500 \times 6 - 250 \times 6^2 = K19,000$. The cost is not twice as much

11. (a) Graph the function (4)

$$y = e - e^x.$$

(b) Use your graph to specify the domain, range intercepts and asymptote of $y = e - e^x$. (5)

**Solution:** $y = -e^x + e$. Domain: $(-\infty, \infty)$, range: $(-\infty, e)$, $y$-intercept: $e - 1$, $x$-intercept: 1, Horizontal asymptote: $y = e$

12. (a) Write the equation $\frac{1}{125} = 5^{-3}$ in logarithmic form (2)

(b) 3] Evaluate $\log_2 8\sqrt{2}$

13. Find the domain of the function (6)

$$y = \ln(2 - x - x^2)$$

**Solution:** Solve the inequality $2 - x - x^2 > 0$. Key numbers are obtained from $(x - 1)(x + 2) = 0$. These are $x = 1$ and $x = 2$

| Interval | Test no. | $x - 1$ | $x + 2$ | $(x - 1)(x + 2)$ |
|---|---|---|---|---|
| $(-\infty, -2)$ | $-3$ | neg. | neg. | pos. |
| $(-2, 1)$ | 0 | neg. | pos. | neg. |
| $(1, \infty)$ | 2 | pos. | pos. | pos. |

Solution is $(-\infty, -2) \cup (1, \infty)$.

14. Prove that (5)

$$\frac{\log_a x}{\log_{ab} x} = 1 + \log_a b$$

**Solution:**

$$\begin{aligned}\frac{\log_a x}{\log_{ab} x} = 1 + \log_a b &= \log_a x \div \frac{\log_a x}{\log_a ab} \\ &= \log_a x \times \frac{\log_a ab}{\log_a x} \\ &= \log_a ab \\ &= \log_a a + \log_a b \\ &= 1 + \log_a b\end{aligned}$$